

Design and Evaluation of a Self-Correcting, Privacy-Preserving Agentic RAG System using Quantized Small Language Models

Course No: AIMLCZG628T

DISSERTATION

by:

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2024AB05158

Degree Program: M.Tech in AI & ML

Research Area:

**Design and Evaluation of a Self-Correcting, Privacy-
Preserving Agentic RAG System using Quantized
Small Language Models**

Dissertation / Project Work carried out at:

Hewlett Packard Enterprise, Bengaluru



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Contents

1. Broad Area of Work	3
2. Background	3
3. Objectives	3
4. Scope of Work	4
5. Plan of Work	4
6. Literature References	5
7. Particulars of the Supervisor and Examiner	6
8. Remarks of the Supervisor	6

1. Broad Area of Work

The current research project focus in the following application areas:

- **Natural language processing (NLP):** Should be capable of understanding user queries and extract meaningful context from the documents that are fed and obtain accurate responses.
- **Information Retrieval (IR):** To fetch the most relevant document chunks using similarity search (Top-K retrieval method).
- **Knowledge representation:** To convert and store document content as vector embeddings for efficient search and retrieval.**Vector Databases:** To store and query high-dimensional embeddings efficiently (e.g., FAISS)**Machine learning:** To generate embeddings, rank relevant content, and improve answer quality using learned patterns.
- **Generative AI (SLMs):** To generate human-like answers grounded in retrieved document context using Small Language Models.

2. Background

In the recent times, advancements in the field of **Retrieval-Augmented Generation (RAG)** has improved the reliability of AI systems by properly grounding responses in the external knowledge sources and also by reducing hallucinations.

Most of the implementations today in real world, rely on large cloud-based language models, raising concerns around **data privacy, cost, and latency**.

In order to address these challenges, this particular project explores the use of **quantized Small Language Models (SLMs)** deployed in a private environment.

Also targets, incorporating an **agentic and self-correcting mechanism** in order to enable the system to iteratively refine responses and improve accuracy.

This aim of the project is to build a **privacy-preserving, efficient, and reliable AI system** suitable for enterprise and on-premise use cases.

3. Objectives

The objectives of my project are as follows:

- To design and develop a **Retrieval-Augmented Generation (RAG) system** that can answer user queries accurately using only relevant information from provided documents.
- To implement a **privacy-preserving architecture** by ensuring all data processing and model inference occur in a local or on-premise environment without relying on external cloud services.
- To integrate **quantized Small Language Models (SLMs)** in order to reduce computational requirements while maintaining acceptable performance and response quality.
- To incorporate a self-correcting mechanism that evaluates generated responses and refines them iteratively to minimize errors and hallucinations.
- To develop an **agentic framework** that enables the system to perform multi-step reasoning, including retrieval, response generation, and validation.

4. Scope of Work

The scope of this project includes designing and implementing a **private, agentic RAG system** using quantized Small Language Models to enable efficient and secure document-based question answering.

It involves building modules for document ingestion, embedding, retrieval, response generation, and a self-correction mechanism within a fully local environment.

The project also covers performance evaluation in terms of accuracy, latency, and effort towards reduction of hallucinations.

5. Plan of Work

Phases	Start Date-End Date	Work to be done
Dissertation Outline	27 April 2026 – 5 May 2026	Literature Review and prepare Dissertation Outline
Design & Development	5 May 2026- 31 May 2026	Design & Development Activity
Testing	31 May 2026 – 15 June 2026	Software Testing, User Evaluation & Conclusion
Dissertation Review	16 June 2026 – 30 June 2026	Submit Dissertation to Supervisor & Additional Examiner for review and feedback
Submission	1 July – 28 July	Final Review and submission of Dissertation

6. Literature Survey

In recent years, the field of Natural Language Processing has seen rapid progress with the development of large language models. However, one of the key limitations of these models is their tendency to generate incorrect or outdated information, commonly referred to as hallucination. To address this issue, researchers have explored techniques that allow models to access external knowledge sources during response generation. One such approach is Retrieval-Augmented Generation (RAG), which combines information retrieval with text generation.

To understand this better, consider a simple example: if a user asks a question about a company policy stored in an internal document, a traditional language model might try to answer based on its training data, which may not include that policy. In contrast, a RAG-based system first retrieves the relevant section of the document and then generates an answer grounded in that content. This makes the response more accurate and context-specific.

The concept of RAG was formally introduced by Lewis et al. (2020), where the authors proposed combining a retriever and a generator to improve performance in knowledge-intensive tasks. This idea builds on earlier work such as REALM by Guu et al. (2020), which demonstrated that models could retrieve relevant information from large corpora during both training and inference, rather than relying solely on stored knowledge. These foundational works established the importance of integrating retrieval mechanisms into language models.

Over time, several studies have analyzed and expanded upon the RAG framework. A systematic literature review conducted in 2025 highlights that most RAG systems follow a pipeline consisting of retrieval, augmentation, and generation. While this architecture improves factual accuracy, the review also points out ongoing challenges such as retrieving irrelevant information and effectively combining multiple pieces of context.

Further evaluation of RAG systems has been carried out by Chen et al. (2023), who showed that although retrieval significantly enhances performance, it does not completely eliminate issues like

hallucination or noise from irrelevant documents. This indicates that simply retrieving information is not sufficient; the system must also be able to assess and refine its own outputs.

To support better evaluation, frameworks such as RAGAS (2024) have been proposed. RAGAS introduces metrics to measure context relevance, answer correctness, and faithfulness without requiring manually labeled data. Similarly, ARES (2024) provides an automated way to evaluate RAG systems using lightweight models, making large-scale testing more practical.

Another important direction in recent research is the use of Small Language Models (SLMs) in combination with RAG. Liu et al. (2024) demonstrate that when supported by an effective retrieval mechanism, SLMs can achieve performance comparable to larger models while requiring significantly fewer computational resources. This is particularly relevant for private or on-premise deployments, where resource constraints and data privacy are major concerns.

Practical applications of RAG can be seen in systems like PaperQA (2023), which focuses on answering questions from scientific documents. Such systems highlight the potential of RAG in domain-specific use cases where accuracy and grounding in source material are critical.

Based on the existing literature, it is evident that while RAG has significantly improved the reliability of language models, there are still gaps to be addressed. These include the need for better handling of irrelevant retrievals, mechanisms for self-correction, and reducing dependence on large models. The proposed project builds on these insights by introducing a self-correcting, agentic RAG system that uses quantized small language models in a privacy-preserving environment, aiming to improve both efficiency and reliability.

7. Literature References

The state art is the base for any successful research project. In current project, the literature inclined towards the Retrieval-Augmented Generation (RAG) is considered.. The following are referred journals from the preliminary literature review.

[1] Lewis et al., 2020 – Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks. Introduced the RAG framework combining retrieval + generation

[2] Guu et al., 2020 – REALM: Retrieval-Augmented Language Model Pre-training

[3] Systematic Literature Review of RAG (2025)

[4] Chen et al., 2023 – Benchmarking LLMs in RAG

[5] RAGAS (2024) – Evaluation framework for RAG

[6] Liu et al., 2024 – Can Small Language Models with RAG Replace LLMs?

[7] PaperQA (2023) – Scientific document QA using RAG

8. Particulars of the Supervisor and Examiner

	Supervisor	Additional Examiner
Name	Saravanan Chidambaram	Rama Krishna Gurram
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9. Remarks of the Supervisor

The proposal and abstract are clear and well-defined. Everything looks good. Please continue with the work.

Information about the Supervisor:

BIRLA INSTITUTE OF TECHNOLOGY & SCIENCE, PILANI
WORK INTEGRATED LEARNING PROGRAMMES (WILP) DIVISION
SECOND SEMESTER OF ACADEMIC YEAR 2021-2022

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DISSERTATION / PROJECT / PROJECT WORK TITLE	Design and Evaluation of a Self-Correcting, Privacy-Preserving Agentic RAG System using Quantized Small Language Models

Please prepare the outline as a separate document with the following sections along with the above identification information.

1. Cover Page with Student ID No., Name, Course Number, Course Title and Dissertation / Project / Project Work Title, Broad Academic Area of Work.
 2. Background (Relevance of the Project to the current work environment in the employing organization)
 3. Objectives
 4. Scope of Work (To be done by the student independently)
 5. Plan of Work (Work to be done during the semester)
 6. Literature References
 7. Particulars of the Supervisor and Additional Examiner
 8. Remarks of the Supervisor
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Signature of Student	Signature of Supervisor	Signature of Additional Examiner
Name: Kathula Deepak	Name: Saravanan Chidambaram	Name: Rama Gurram